

Quantum Computing — At a Glance

Qubit & Superposition Qubit: $\alpha 0\rangle + \beta 1\rangle$ — superposition enables parallelism. $ \alpha ^2 + \beta ^2 = 1$	Entanglement Qubits become strongly correlated. Example Bell state: $(00\rangle + 11\rangle)/\sqrt{2}$ — enables teleportation, QKD, networks.	Measurement Measure $ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$ → $P(0)= \alpha ^2$, $P(1)= \beta ^2$. State collapses to $ 0\rangle$ or $ 1\rangle$.	No-Cloning & Security Unknown states cannot be perfectly copied — foundation for QKD and secure quantum comms.	Matrices & Gates Unitary: $U^\dagger U = I$ (reversible gates). Hermitian: $H^\dagger = H$ (observables). Examples: I, X, Y, Z, H.	Key Gates I (identity), X (bit flip), Y (bit+phase), Z (phase flip), H (creates superposition).
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Foundations

- Quantum info uses quantum mechanics
- Computational basis: $|0\rangle = [1\ 0]^T$, $|1\rangle = [0\ 1]^T$
- Hilbert space: complete vector space with inner product

Advantages & Limits

- Quantum parallelism → speedups (Shor, Grover)
- Cannot clone unknown states (No-Cloning theorem)
- Measurement destroys superposition

Linear Algebra Notes

- Inner product $\langle\phi|\psi\rangle$ — positivity, linearity, conjugate symmetry
- Outer product $|\psi\rangle\langle\phi|$
- Normal matrices: $N N^\dagger = N^\dagger N$

Applications

- Quantum Cryptography / QKD
- Quantum Algorithms (Shor, Grover)
- Drug discovery, materials, optimization, finance, AI, simulation

Core Equations $ \psi\rangle = \alpha 0\rangle + \beta 1\rangle \cdot \alpha ^2 + \beta ^2 = 1 \cdot U^\dagger U = I \cdot H^\dagger = H$	Classical vs Quantum Classical: bits, copy freely, sequential. Quantum: qubits, no cloning, superposition, entanglement, parallelism.	Quick Visuals Bloch sphere • Bell pair • Gate matrices • Measurement collapse
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■ **Study Tip:** Keep slides concise — speak to equations, gates, and applications. Use visuals above as anchors.